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absorption through the xylem portion of the twig. Thus, the rate of water absorption is equal to the rate of transpiration.

2. Cobalt chloride ($CoCl_2$) paper method

Select a healthy dorsiventral leaf and clean its upper and lower surface with dry cotton. Now place a dry Cobalt chloride (**$CoCl_2$**) strips on both surface and immediately cover the paper with glass slides and immobilize them. It will be observed after some time that the $CoCl_2$ strip of lower epidermis turns pink. This indicates that $CoCl_2$ becomes hydrated ($CoCl_2 \cdot 2H_2O$ or $CoCl_2 \cdot 4H_2O$) due to water vapours coming out through stomata. The rate of transpiration is more on the lower surface than in the upper surface of the dorsiventral leaf.

11.6.8 Significance of transpiration

Transpiration leads to loss of water, as stated earlier in this lesson 95% of absorbed water is lost in transpiration. It seems to be an evil process to plants. However, number of process like absorption of water, ascent of sap and mineral absorption directly relay on the transpiration. Moreover plants withstand against scorching sunlight due to transpiration. Hence the transpiration is a "**necessary evil**" as stated by **Curtis**.

11.7 Translocation of Organic Solutes

Leaves synthesize food material through photosynthesis and store in the form of starch grains. When required the starch is converted into simple sugars. They must be transported to various parts of the plant system for further utilization. However, the site of food production (leaves) and site of utilization are separated far apart. Hence, the organic food has to be transported to these areas.

The phenomenon of food transportation from the site of synthesis to the site of utilization is known as **translocation of organic solutes**. The term **solute** denotes food material that moves in a solution.

11.7.1 Path of Translocation

It has now been well established that phloem is the path of translocation of solutes. Ringing or girdling experiment will clearly demonstrate the translocation of solute by phloem.

11.7.2 Ringing or girdling experiment

The experiment involves the removal of all the tissue outside to vascular cambium (bark, cortex, and phloem) in woody stems except xylem. Xylem is the only remaining tissue in the girdled area which connects upper and lower part of the plant. This setup is placed in a beaker of water. After some time, it is observed that a swelling on the upper part of the ring appears as a result of the accumulation of food material (Figure 11.20). If the experiment continues within days, the roots die first. It is because, the supply of food material to the root is cut down by the removal of phloem. The roots cannot synthesize their food and so they die first. As the roots gradually die the upper part (stem), which depends on root for the ascent of sap, will ultimately die.

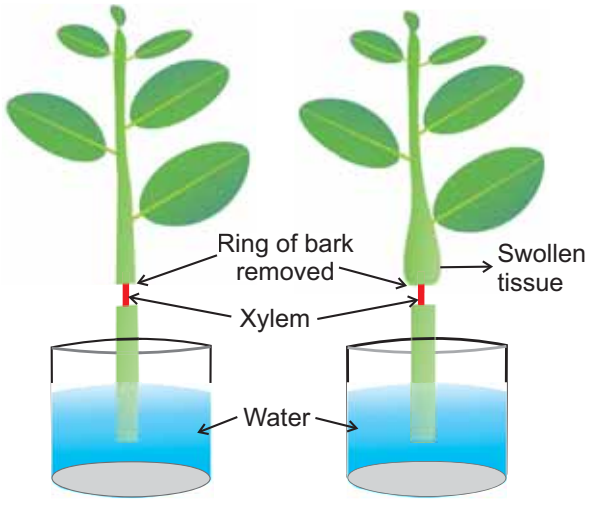


Figure 11.20: Ringing experiment